



IMDb Movie Rating Prediction Model

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Introduction

Our objective with the 2026 IMDb Prediction Challenge is to develop a statistical model capable of predicting the IMDb ratings of twelve upcoming blockbuster films using information from a dataset of approximately 2000 released films. Given that the IMDb ratings are commonly used as an indication of audience/general industry response to a film, we find that they are a reliable measure of film performance. With that in mind, we aim to construct the most accurate predictive model that could estimate the expected ratings of these twelve future film releases.

The dataset we used for this challenge included a variety of variables which described the film itself on a structural and creative level (duration, genre, language, etc.). These variables also described the resources and marketing behind the film (budget, number of news articles, etc.), as well as the popularity or experience of the actors involved (number of well-known actors, actor popularity scores, etc). With these variables in mind, we would apply statistical modeling techniques to identify which variables are most strongly correlated with IMDb ratings. But predicting IMDb ratings clearly comes with several challenges such as audience subjectivity, many diverse preferences, different cultural factors, and more. Which is why our model will balance complexity and generalization to really capture meaningful predictions while avoiding risk of overfitting.

Our report will essentially evaluate multiple statistical models and select the final model based on the best predictive performance and its ability to generalize unseen data. We will then use the final selected model to generate predicted IMDb ratings for the twelve upcoming films.

Data Description

For this research, the IMDb score represents the dependent variable. The independent variables were separated into 3 distinct predictor categories. First, there is a group of variables that capture production characteristics, including the duration, country, budget, and production company that go into movie making and structuring. There are also popularity and public receptions, found through media coverage (number of news articles) and actors' individual scores. Finally, the content category related to a movie's narrative includes variables such as maturity rating, genres and summative keywords.

The original dataset contained approximately 1,930 movies. To ensure greater compatibility with the test sample of 12 movies, several filters were applied. First, movies produced only in countries present in our test set, the United States and United Kingdom, were retained. As movie budget and production vary across countries, narrowing the selection of areas would provide more stability for our model. We also restricted the dataset to colored movies, commonly found in all 12 test movies. We removed certain genres such as western, sports and war, as they did not appear in the test sample, thus not providing explanatory power. We also narrowed the releases exclusively to the winter period, from January to April, avoiding seasonal influx present during Christmas or the summertime, which could have introduced bias in the dataset. This filtering left us with 477 movies used as our final practice set.

The simple linear regressions depicted several patterns about the structure of the cleaned dataset. Indeed, the distribution of movie ratings has a mean of 6.5 points, depicting a slightly skewed distribution (see Figure 1), with a few movies receiving extreme ratings. Independent variables such as the number articles and actors' star meters exhibit strong right-skewness (see Figure 2),

suggesting only a slight number of movies receive strong media coverage and experience high public interest, potentially due to the influence of a few controversial movies. This is also highlighted with a huge gap between the value of 844 articles, the third quartile, as opposed to a maximum of 8848 (see Figure 3). Similarly, movies with globally known actors also depict strong skewness in the data set, resulting from a few “superstars” outliers (see Figure 4)

The relationship between variables also provides insight into the factors associated with movie ratings. For instance, longer movies tend to receive higher IMDb ratings, which reflects that longer runtimes might be more critically appreciated. Greater media coverage, through a number of news articles, is also associated with higher ratings. Stronger exposure would indicate greater quality or cultural relevance. Furthermore, recent movies have a slight negative correlation with ratings, whereas older films are more established and recognized, thus experiencing higher scores.

As for the categorical variables, movies from the USA have a score of 0.63 lower than UK movies, with 3.5% IMDb score explained by the variation, playing a minor role in rating. PG and PG-13 rated movies both experience a drop in rating (-0.30 and -0.42 respectively) compared to R-rated movies. This could suggest that PG and PG-13 audiences target a wider public with diverse preferences, whereas R-rated movies are restricted to more mature viewers who may be more accustomed to the content. Notably, both cinematographer and main actor variables have over 300 unique entries; it becomes harder to capture consistent patterns and their effects on IMDb ratings. The insufficient repetitions may reflect the long time span covered by the dataset, involving a wide range of professionals involved in movies released over decades. Similarly, some large studios, such as Warner Bros, Universal and Paramount produce over a dozen movies in the dataset. This is commonly associated with larger budget and exposure but impact cannot be properly assessed with ratings due to the large number of small studios and wide variability.

We can observe moderate overlapping collinearity in genres such as thriller and crime having a strong overlap of 0.42. Similarly, thriller and action also have an overlap (0.39) as thrillers tend to incorporate action segments, increasing suspense. The duration variable also depicted a collinearity of 0.36 as dramas could have longer narratives. We can assess that genres are not fully independent from one another, with characteristics naturally overlapping in storytelling.

Nonetheless, we can notice low collinearity between variables in the popularity category, as each covers different aspects. For instance, actor star meters cover the individual popularity scale whereas IMDb movie meter touches on movie searching and visibility. Moreover, actors 1 and actor 2 scores have a low collinearity of 0.12, implying that the popularity of the main actor does not strongly predict the score of another actor in the cast.

Overall, several variables, considered independently, describe a small portion of the dataset variation in IMDb ratings. Several predictors exhibit moderate to high skewness in their distributions, often influenced by outliers. Large budget allocation may be linked to an increase in quality and greater exposure. However, strong variability remains across categorical variables, limiting the predictability of ratings.

Model Selection

1. Overall Methodology and Strategy

The main goal for this project is to predict the IMDb score of the 12 upcoming 2026 blockbuster releases. In order to achieve this, our thought process was to focus on finding the optimal balance between model flexibility and out-of-sample predictive power, not by strictly forcing higher R2.

2. Rationale for Including and Excluding Predictors

The initial phase required a strict filtering of variables to prevent structural overfitting. Categorical variables with an excessive number of unique observations such as actor1, actor2, actor3, plot_keywords, director, distributor, and cinematographer, were excluded. Because it appeared extremely rarely, we would need to generate dummy variables, causing overfitting.

Based on our correlation analysis, duration showed the strongest positive relationship with IMDb scores ($r=0.40$), indicating that higher film ratings tend to be longer. Similarly, drama ($r=0.34$) and nb_news_articles ($r=0.27$) exhibited positive correlations, while variables like release_year ($r=-0.23$) and nb_faces ($r=-0.22$) were included to show negative declining pressures on ratings.

3. Addressing Model Issues: Collinearity and Heteroskedasticity

We utilized a correlation matrix to verify that the independent variables were not providing redundant information (see Figure 9). Fortunately, no severe collinearity was detected. No two predictors have a correlation coefficient higher than 0.5. Previously stated, moderate correlations existed, which fall well below the standard danger thresholds.

The tests we conducted revealed that there is heteroskedasticity: duration ($p=0.0013$), movie_budget ($p=0.0039$), and movie_meter_IMDbpro ($p=0.0041$). To correct for the heteroskedasticity, we used logarithmic transformations for heavily skewed numeric predictors, to decrease variance and transform outliers closer to a normal distribution.

4. Rationale for Non-Linear Functional Forms (Linear vs. Quadratic vs. Splines)

From all our previous tests, the non-linear model fits. To maximize predictive accuracy, we utilized our strongest numerical predictor, duration, in a rigorous non-linear optimization process.

Plotting IMDb score against duration with a flexible Locally Estimated Scatterplot Smoothing (LOESS) curve revealed a non-linear relationship (see Figure 5). Scores increase steadily with runtime but peak at approximately 225 minutes before bending downwards. Forcing a linear fit to the model would drastically decrease the accuracy of the prediction

To determine the exact mathematical shape, we first tested Polynomial functions using a Leave-One-Out Cross-Validation (LOOCV) loop for degrees 1 through 5 (see Figure 6). While the Degree 2 (quadratic) polynomial yielded a respectable out-of-sample Mean Squared Error (MSE) of 0.9487, moving to a Degree 3 (cubic) polynomial caused a massive and catastrophic jump in the MSE to 2.1345. This is a clear sign of overfitting; higher-degree polynomials become highly volatile and vary wildly at the tails of the data, which would result in absurd predictions for extremely short or long movies.

To capture the curve safely without the tail-end volatility of polynomials, we modeled duration using Regression Splines. Because we did not have prior institutional knowledge detailing exact "minute-marks" where audience grading shifts, we relied on statistical percentiles. We placed two knots uniformly at the 33rd and 66th percentiles of the duration distribution to divide the data evenly into three segments.

We then executed a 10-Fold Cross-Validation loop testing Spline degrees 1 through 4 (see Figure 7). The Degree 3 (Cubic) Spline produced the absolute lowest out-of-sample error across all tests (MSE = 0.9116). Unlike the polynomial models, the spline remained mathematically stable across its segments.

As a final confirmation, we ran an Analysis of Variance (ANOVA) comparing the basic linear model to our newly constructed cubic spline model (see Figure 8). The test yielded a highly significant p-value of 1.937×10^{-8} ($p < 0.05$). This provided the final proof that not using the linear assumption in favor of a 3rd-degree spline with two percentile knots statistically improved the model's fit. We applied this exact cross-validation methodology to our other continuous variables, allowing practical out-of-sample MSE tests to determine our final model.

Results - Building and testing our first complete model

We chose our features through observing the correlation of different features to the IMDB_Score and evaluating their strength, we chose to be quite critical with our features and set a p-value threshold of 0.05. Based on the features left when we applied this model we then checked the value of their coefficients to see which had the biggest sway in IMDB_score which brought us to the results shown in figure 9.

The equation initially selected in the model selection process employed features sparingly as to avoid overfitting and to have a more cohesive model, having only duration, drama, news articles and nb_faces as features. In the end we included a few more features as we found this model was too restrictive and showed signs of underfitting, so we incorporated additional genre features to improve fit. Additionally we experimented with interactions between variables, using intuition initially to estimate which features would work well together then confirming them statistically by reviewing their metrics. This led us to get the model in exhibit 1, as we believed the highest rated films would be longer dramas with a lot of media buzz, these films particularly would signify films with award buzz which would be sure to receive good IMDB scores.

We intentionally omitted a few predictors with strong correlation to our target (see Figure 10) variable as we believed they were not conducive to our objective of predicting 2025 blockbusters. Additionally we wanted to avoid overfitting and as such wanted to limit the included features. One such example is release_year, despite showing a strong correlation in our initial test the release year wouldn't be relevant to our project as all the movies we predicted for were released in 2025. Furthermore, the correlation between release_year and IMDB_score is somewhat spurious as release_year has no logical change to a movie's production and quality and often reflects a nostalgic augmentation in score or a selective practice about which films to review (ie. older films of poor quality are often forgotten, while great ones are remembered).

We also omitted the budget field as per the data dictionary "half of films don't have publicly available budgets" and we were concerned this would affect the integrity of the data. Furthermore, all the movies are stated to be "blockbusters" which means they are all relatively high in budget value and thus would neutralize the presence of budgets in differentiating their scores.

We applied a 10-fold cross-validation method to estimate the predictive ability of our model and to limit the risk of overfitting. This measure allows us to divide the dataset into ten different subsets and train the model based on nine subsets, while testing on the singular remaining subset ten

different times so that each observation can be used once as validation data. This allowed us to accurately approximate the mean squared error (MSE) on unseen data and produced a more definitive estimate of predictive power rather than simply evaluating the model on the training dataset.

The model that was selected after our initial review (see Figure 11) displayed a cross-validated MSE of approximately 0.81 with an adjusted R-squared of 0.396 (see Table 1), suggesting that 39.6% of the variation in IMDb scores can be explained by the predictors that were included in the model. This result is rather fitting since it can be quite challenging to capture all predictors for an opinionated and moderately subjective metric such as IMDb score, knowing that there exists a large number of unobserved factors that can greatly vary from person to person.

While predicting the IMDb scores for the test dataset, we encountered an issue related to the “country” variable, where the “Philippines” was present in our test dataset, but absent in the training dataset. This phenomenon caused errors while running predictions, since it is necessary for regression models containing categorical predictors to have all attributes present during training. We decided to remove the “country” variable from our model to address this issue and maintain consistency throughout our predictions, which resulted in a marginal decrease of adjusted R-squared (from 0.407 to 0.396) and an increase in MSE (from 0.785 to 0.812). This relatively small change in adjusted R-squared and MSE implies that the predictor did not add a significant amount of predictive value to the model and withdrawing it from the final model is a reasonable and justifiable approach in order to address the issue it created.

Applying the final model (see Figure 12) to the test dataset containing the twelve upcoming blockbuster films with all their respective details, we generated predictions for their IMDb scores (see Table 2). Our predictions all fall between 5.3-6.8, exhibiting a mean of 6.43 and a median of 6.5. Furthermore, just over half the films in the training set, or 51.3%, had an IMDb score between the same interval, which suggests that our predictions on the training set are quite consistent with the distribution of the IMDb scores in the training dataset. This observation suggests that the model is generating plausible predictions for unseen movies, instead of producing extreme values on either end.

Among the predicted movies, “Snow White” and “The Alto Knights” received the highest predicted ratings, which is directly reflected in the number of news articles written about both movies, which is among the highest of the 12 films, and also the duration with both films being the only ones exceeding the 2 hour mark. On the contrary, “Locked” and “High Rollers” ended with lower predicted values, in which both films had very few articles written about them and were both non-dramas (horror and action) respectively. As table 4 shows the non-drama movies particularly in the action and horror genres tended to receive lower scores whereas the drama movies had slightly higher scores with the notable exception of “The Woman in the Yard” due to its short runtime and low media recognition.

Despite the fact that the model does indeed provide useful predictions, it is essential to acknowledge its limitations. IMDb ratings are strongly influenced by many highly subjective factors, such as personal expectations for a film, cultural trends during the time period the movie premiered, individual bias for directors, genres, actors, etc. that are simply not entirely captured and represented amidst the dataset, resulting in a limitation in predictive power. Conversely, the 10-K-fold cross-validation model outputs relatively stable and reasonably justifiable results and

out-of-sample performance, making it, at the very least, moderately appropriate at predicting the IMDb score for films.

Appendix

Figure 1 - Histogram of IMDb Score

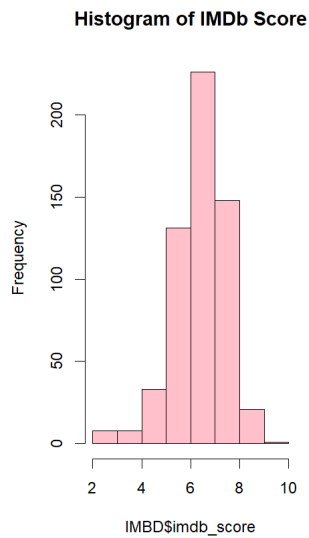


Figure 2 - Histogram of Actor 1 Star Meter

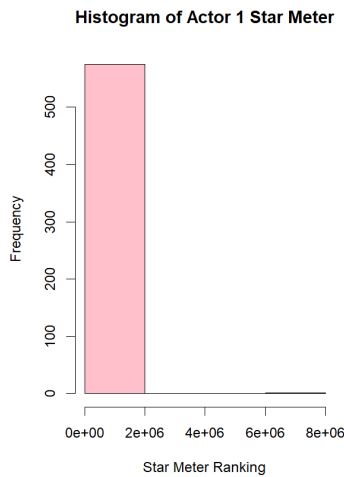


Figure 3 - Box plot of Number of News Articles

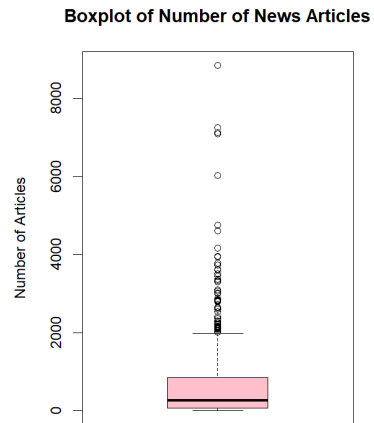


Figure 4 - Boxplot of Actor 2 Star Meter

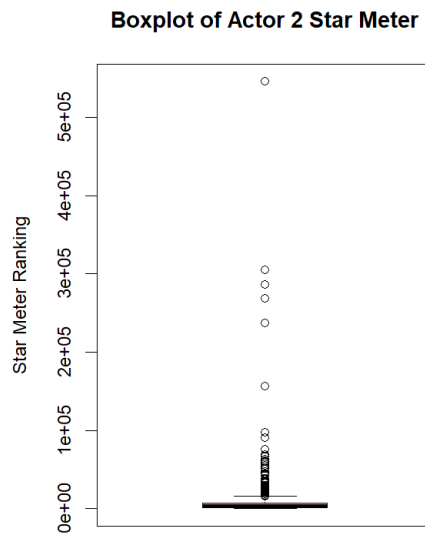


Figure 5 - Visual plot of IMDb Score vs Duration with Loess Curve

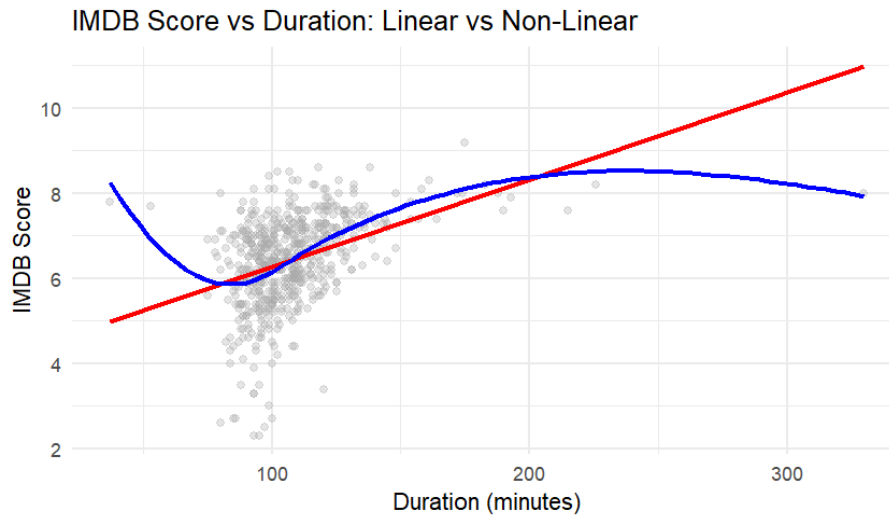


Figure 6 - LOOCV's MSE Test Results Degree 1-5

	Degree	Out_of_Sample_MSE
1	1	0.9584
2	2	0.9487
3	3	2.1345
4	4	2.1786
5	5	7.3253

Figure 7 - B-Spline's MSE Test Results Degree 1-4

	Spline_Degree	KFold_MSE
1	1	0.9285
2	2	0.9431
3	3	0.9116
4	4	1.1442

Figure 8 - ANOVA Test Results of Linear vs Winning Non-linear Model

Analysis of Variance Table

Model 1: imdb_score ~ duration

Model 2: imdb_score ~ bs(duration, knots = knots_dur, degree = 3)

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	573	542.50				
2	569	504.24	4	38.26	10.794	1.937e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Figure 9 - Correlation Matrix

	imdb_score	movie_budget		duration	aspect_ratio		actor1_star_meter
imdb_score	1.00	-0.08	imdb_score	0.40	-0.02	imdb_score	0.10
movie_budget	-0.08	1.00	movie_budget	0.19	0.22	movie_budget	-0.06
release_day	-0.05	-0.02	release_day	-0.03	-0.07	release_day	-0.07
release_year	-0.23	0.15	release_year	-0.25	0.32	release_year	-0.24
duration	0.40	0.19	duration	1.00	0.06	duration	0.01
aspect_ratio	-0.02	0.22	aspect_ratio	0.06	1.00	aspect_ratio	-0.11
nb_news_articles	0.27	0.02	nb_news_articles	0.11	0.00	nb_news_articles	-0.01
actor1_star_meter	0.10	-0.06	actor1_star_meter	0.01	-0.11	actor1_star_meter	1.00
actor2_star_meter	0.03	-0.10	actor2_star_meter	-0.09	0.02	actor2_star_meter	0.14
actor3_star_meter	-0.02	0.01	actor3_star_meter	-0.04	-0.07	actor3_star_meter	0.02
nb_faces	-0.22	0.07	nb_faces	-0.06	-0.03	nb_faces	-0.04
action	-0.15	0.25	action	-0.04	0.19	action	-0.02
adventure	-0.06	0.12	adventure	-0.02	0.03	adventure	-0.01
scifi	-0.08	0.10	scifi	-0.07	0.01	scifi	-0.01
thriller	-0.07	0.20	thriller	-0.04	0.20	thriller	-0.04
musical	-0.03	-0.06	musical	0.02	-0.08	musical	0.03
romance	-0.02	0.01	romance	-0.02	-0.12	romance	0.06
horror	-0.15	-0.07	horror	-0.16	0.01	horror	-0.02
drama	0.34	-0.14	drama	0.36	0.11	drama	0.02
animation	0.02	0.13	animation	-0.11	-0.05	animation	-0.01
crime	0.06	0.08	crime	0.06	0.09	crime	-0.03
movie_meter_IMDBpro	-0.18	-0.16	movie_meter_IMDBpro	-0.13	0.00	movie_meter_IMDBpro	-0.02
	release_day	release_year		nb_news_articles			actor2_star_meter
imdb_score	-0.05	-0.23	imdb_score	0.27		imdb_score	0.03
movie_budget	-0.02	0.15	movie_budget	0.02		movie_budget	-0.10
release_day	1.00	0.07	release_day	0.03		release_day	-0.01
release_year	0.07	1.00	release_year	0.16		release_year	0.07
duration	-0.03	-0.25	duration	0.11		duration	-0.09
aspect_ratio	-0.07	0.32	aspect_ratio	0.00		aspect_ratio	0.02
nb_news_articles	0.03	0.16	nb_news_articles	1.00		nb_news_articles	-0.01
actor1_star_meter	-0.07	-0.24	actor1_star_meter	-0.01		actor1_star_meter	0.14
actor2_star_meter	-0.01	0.07	actor2_star_meter	-0.01		actor2_star_meter	1.00
actor3_star_meter	0.02	-0.03	actor3_star_meter	-0.05		actor3_star_meter	0.11
nb_faces	0.01	0.11	nb_faces	-0.06		nb_faces	-0.08
action	0.07	0.07	action	0.05		action	0.02
adventure	-0.01	-0.08	adventure	-0.10		adventure	-0.03
scifi	0.04	0.01	scifi	0.08		scifi	0.01
thriller	-0.04	0.08	thriller	-0.04		thriller	-0.02
musical	-0.03	-0.06	musical	0.01		musical	0.10
romance	-0.02	0.00	romance	0.03		romance	-0.05
horror	0.00	-0.04	horror	0.08		horror	0.00
drama	-0.05	0.00	drama	0.03		drama	-0.07
animation	0.02	0.04	animation	-0.05		animation	-0.01
crime	0.03	0.02	crime	-0.07		crime	0.02
movie_meter_IMDBpro	0.03	0.07	movie_meter_IMDBpro	-0.20		movie_meter_IMDBpro	0.15

$$\hat{y} = B_1^3(\text{duration}) \times B_2^2(\text{drama}) + B_3^2(\text{nb_news_articles}) + B_4^3(\text{nb_faces}) + B_5^3(\text{action}) + B_6^3(\text{horror}) + B_7^1(\text{movie_meter_IMDBpro}) + \varepsilon$$

Table 1 - Adjusted R-squared and Mean MSE for all Regression Models

	Adjusted R-squared	Mean Mse
Model 4	0.3726	3.27754
Model 7	0.3655	0.8215602
Model A	0.3726	3.27754
Model B	0.3746	0.815394
Model C	0.4091	4.954786
Model D	0.4109	0.8114886
Model E	0.3963	0.8119031
Model F	0.436	10.0367(fluctuating)
Model G	0.4225	3.276942 (fluctuating)

Table 2 - Predicted IMDb Score for each Movie in the Test Dataset

Movie Title	Predicted IMDb Score
O'Dessa	6.04
Black Bag	5.67
High Rollers	5.43
Novocaine	5.71
The Day the Earth Blew Up	5.80
Ash	5.44
Locked	5.24

Snow White	6.77
The Alto Knights	6.52
A Working Man	5.90
My Love Will Make you Disappear	6.19
The Woman in the Yard	5.40

Table 3- Films with their genres and scores

Movie Title	Genres	Predicted IMDb Score
O'Dessa	Musical-Drama	6.04
Black Bag	Thriller-Drama	5.67
High Rollers	Action	5.43
Novocaine	Action-Thriller	5.71
The Day the Earth Blew Up	Adventure-Animation	5.80
Ash	Scifi-Thriller-Horror	5.44
Locked	Thriller-Horror	5.24
Snow White	Adventure-	6.77
The Alto Knights	Drama-Crime	6.52
A Working Man	Action-Thriller	5.90
My Love Will Make you Disappear	Romance	6.19
The Woman in the Yard	Horror-Drama	5.40